Class tests : Best two of three
weight : 10 + 10

Midterm : 20

Semester : 60

Generating fn:

Sp. you have a seq^n $\{a_n : n \geq 0\}$

Define $A(s) = \sum_{n=0}^{\infty} a_n s^n$

R = rad. of conv. = $\dfrac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}$

Note, if $|a_n| \leq M$ $\forall n \geq 0$,

then $R \geq 1$.

Example :-

$a_n = 1 \quad \forall\, n \geq 0,$

$$A(s) = \frac{1}{1-s}, \quad |s| < 1$$

$$\left(\sum_{n=0}^{\infty} s^n \right)' = \sum_{n=0}^{\infty} n s^{n-1}$$

$a_n = n,$

$$A(s) = \frac{1}{(1-s)^2} \times s, \quad |s| < 1$$

Prob. gen. fn (pgf)

X taking values in $\{0, 1, \ldots\}$

$$P_X(s) = \sum_{n=0}^{\infty} P(X=n)\, s^n$$

$$\text{— gen. fn of } \{\overline{\underbrace{P(X=n)}_{=p_n}}\}$$

Properties :-

- $P_X(s)$ is inf. diff. on $|s|<1$.

$$P_X^{(k)}(s) = \sum_{n=0}^{\infty} n(n-1)\cdots(n-k+1)\, p_n s^{n-k}$$

- Put $s=0$, $P^{(k)}(0) = p_k \cdot k!$
- Distr of X is determined by P_X

Lemma: (Abel's lemma)

a) If $\sum_{n=0}^{\infty} a_n < \infty$, then

$$\lim_{s \uparrow 1} A(s) = \sum_{n=0}^{\infty} a_n$$

b) If $a_n \geq 0$, then

$$\lim_{s \uparrow 1} A(s) = \sum_{n=0}^{\infty} a_n$$

$$\lim_{s \uparrow 1} P_X(s) = \sum_{n=0}^{\infty} p_n = 1 \quad \Big| \quad P_X(1) = 1$$

$$E\big(X(X-1)(X-2)\cdots(X-k+1)\big)$$

$$= \sum_{n=0}^{\infty} n(n-1)(n-2)\cdots(n-k+1)\, p_n$$

$$= \lim_{s \uparrow 1} P^{(k)}(s) \quad \Big| \quad E(X) = \lim_{s \uparrow 1} P_X^{(1)}(s)$$

$$= P_X^{(1)}(1)$$

- Concentrate on [0,1]

- $$P'(s) = \sum_{n=0}^{\infty} n p_n s^{n-1} \geq 0$$

  > 0 if $p_0 < 1$

  incr. (strictly if $p_0 < 1$).

- $$P''(s) = \sum_{n=0}^{\infty} n(n-1) p_n s^{n-2} \geq 0$$

  is convex (strictly if $p_0 + p_1 < 1$)

$$q_n = \mathbb{P}(X > n)$$

$$Q(s) = \sum_{n=0}^{\infty} q_n s^n$$

$$q_n - q_{n+1} = \mathbb{P}(X > n) - \mathbb{P}(X > n+1)$$

$$= \mathbb{P}(X = n+1) \quad \forall\, n \geq 0.$$

$$= p_{n+1}$$

$$q_n s^{n+1} - q_{n+1} s^{n+1} = p_{n+1} s^{n+1}$$

$\Rightarrow$ Sum over n,

$$\sum_{n=0}^{\infty} q_n s^{n+1} - \sum_{n=0}^{\infty} q_{n+1} s^{n+1} = \sum_{n=0}^{\infty} p_{n+1} s^{n+1}$$

$$\Rightarrow s\,Q(s) - \left(Q(s) - q_0\right) = P(s) - p_0$$

$$\Rightarrow Q(s)(s-1) = P(s) - p_0 - q_0$$

$$\boxed{\Rightarrow Q(s) = \frac{1-P(s)}{1-s}} = P(s) - 1.$$

$$P(X=0) \qquad P(X>0)$$

$Q(s) = P'(c_s)$ where $c_s \in (s,1)$.

As $s \uparrow 1$, $c_s \uparrow 1$.

$$\Rightarrow \lim_{s \uparrow 1} Q(s) = \lim_{s \uparrow 1} P'(c_s) = P'(1) = E(X)$$

$$\Rightarrow Q(1) = \lim_{s \uparrow 1} Q(s) = \sum_{n=0}^{\infty} q_n$$

$$\Rightarrow \boxed{E(X) = \sum_{n=0}^{\infty} P(X>n)}$$

$$\boxed{Q(1)}$$